

Chapter - 4

Analyzing Functional Dependencies, Redundancy And Data Anomalies

4.1 Introduction:

The **Nykaa Beauty Store Management System** database is designed to efficiently manage customer information, beauty products, brands, categories, orders, payments, reviews, and shipment details. Functional dependencies help identify the relationships between attributes within each table, ensuring that every non-key attribute depends on the primary key. Redundancy analysis and data anomaly detection help reduce duplicate data, maintain data consistency, and improve database performance. A well-structured database ensures data integrity, minimizes storage redundancy, and supports efficient management of the Nykaa Beauty Store Management System.

1. CUSTOMER TABLE

Functional Dependency:

Customer_ID → Customer_Name, Email, Phone_Number, Address, Password

Redundancy Analysis:

Customer details are stored only once in the Customer table. Order and Review tables reference the customer using Customer_ID, avoiding duplication.

Data Anomalies:

No data anomalies were identified.

Reason: Customer information is stored only once and uniquely identified by Customer_ID. The normalized design prevents insert, update, and delete anomalies.

2. BRAND TABLE

Functional Dependency:

Brand_ID → Brand_Name, Brand_Description

Redundancy Analysis:

Brand information is stored only once. Products reference the brand using Brand_ID, reducing redundancy.

Data Anomalies:

No data anomalies were identified.

Reason: Brand information is stored separately and uniquely identified by Brand_ID. The normalized design prevents insert, update, and delete anomalies.

3. CATEGORY TABLE

Functional Dependency:

Category_ID → Category_Name, Category_Description

Redundancy Analysis:

Each category is stored only once. Products reference the category using Category_ID, avoiding repeated category details.

Data Anomalies:

No data anomalies were identified.

Reason: Category information is stored only once and uniquely identified by Category_ID. The normalized design prevents insert, update, and delete anomalies.

4. PRODUCT TABLE

Functional Dependency:

Product_ID → Product_Name, Brand_ID, Category_ID, Price, Stock, Product_Description

Redundancy Analysis:

Product details are stored only once. Brand and Category are referenced using foreign keys, minimizing duplication.

Data Anomalies:

No data anomalies were identified.

Reason: Product information is uniquely identified by Product_ID and references Brand and Category using foreign keys. The normalized design prevents insert, update, and delete anomalies.

5. ORDER TABLE

Functional Dependency:

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis:

Each order is stored once and linked to a customer using Customer_ID.

Data Anomalies:

No data anomalies were identified.

Reason: Each order is uniquely identified by Order_ID and linked to the Customer table through a foreign key. The normalized design prevents insert, update, and delete anomalies.

6. ORDER ITEM TABLE**Functional Dependency:**

Order_Item_ID → Order_ID, Product_ID, Quantity, Unit_Price

Redundancy Analysis:

Each order item stores only product and quantity details for a specific order, reducing duplicate records.

Data Anomalies:

No data anomalies were identified.

Reason: Each order item is uniquely identified by Order_Item_ID and maintains proper relationships with the Order and Product tables. The normalized design prevents insert, update, and delete anomalies.

7. PAYMENT TABLE**Functional Dependency:**

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Amount

Redundancy Analysis:

Payment information is stored separately and linked to the order using Order_ID.

Data Anomalies:

No data anomalies were identified.

Reason: Payment information is uniquely identified by Payment_ID and linked to the Order table using Order_ID. The normalized design prevents insert, update, and delete anomalies.

8. REVIEW TABLE

Functional Dependency:

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis:

Reviews are stored once and linked to customers and products using foreign keys.

Data Anomalies:

No data anomalies were identified.

Reason: Each review is uniquely identified by Review_ID and linked to the Customer and Product tables using foreign keys. The normalized design prevents insert, update, and delete anomalies.

9. SHIPMENT TABLE

Functional Dependency:

Shipment_ID → Order_ID, Shipment_Address, Shipment_Status, Shipment_Date, Delivery_Partner

Redundancy Analysis:

Shipment information is stored separately and linked to orders using Order_ID, reducing duplicate shipping records.

Data Anomalies:

No data anomalies were identified.

Reason: Shipment information is uniquely identified by Shipment_ID and linked to the Order table using a foreign key. The normalized design prevents insert, update, and delete anomalies.

4.2 Conclusion:

The functional dependency analysis of the **Nykaa Beauty Store Management System** confirms that each table has a well-defined primary key and appropriate relationships among attributes. Redundancy is minimized through normalization and the use of foreign keys, ensuring that data is stored only once whenever possible. The database design effectively prevents insert, update, and delete anomalies, thereby improving data consistency, integrity, and overall system performance. This structured design provides a reliable and efficient foundation for managing all operations of the Nykaa Beauty Store Management System.